Comprehensive Data Sourcing and Knowledge Graph Architecture for the 'Fessi' Waste Disposal Chatbot: Legal Frameworks, Classification Logic, and Domain-Specific Ontologies

Executive Summary

The digital transformation of waste management within complex institutional settings, such as German universities, necessitates a departure from static, document-based guidelines toward dynamic, semantic knowledge systems. The "Fessi" chatbot project represents a paradigm shift in how students, researchers, and facility managers interact with waste disposal regulations. To function effectively, this system requires a high-fidelity Knowledge Graph (KG) populated with authoritative legal codes, intricate classification logic, and granular, university-specific waste descriptions. This report provides an exhaustive analysis of the data landscape required to build such a system, focusing on the *Abfallverzeichnis-Verordnung* (AVV), the European Waste Catalogue (EWC), and the supporting technical rules that dictate compliance.

The core challenge addressed in this report is the translation of the hierarchical, statute-based AVV into a queryable graph structure that can handle the nuance of "mirror entries"—waste types that may be hazardous or non-hazardous depending on their specific chemical composition. The analysis identifies and evaluates primary data sources, ranging from federal legal repositories (*Gesetze-im-Internet*) and European statistical databases (Eurostat) to specialized technical guidelines (TRGS 201, LAGA M18) and unstructured university "Abfall-ABC" directories. Furthermore, it details the logic required to map common university waste streams—such as cytotoxic laboratory residues, electronic components, and construction debris—to their correct six-digit AVV codes. By synthesizing these diverse data streams, the proposed architecture for "Fessi" ensures legal compliance, operational efficiency, and enhanced user engagement in campus sustainability initiatives.

1. The Regulatory Architecture of Waste Classification

The foundation of the "Fessi" knowledge graph lies in a deep understanding of the regulatory framework governing waste in Germany and the European Union. Unlike simple categorization schemes, the legal classification of waste is a rigid, hierarchical process derived from the *Kreislaufwirtschaftsgesetz* (KrWG) and the *Abfallverzeichnis-Verordnung* (AVV). Understanding this architecture is prerequisite to defining the schema of the knowledge

graph.

1.1 The European Waste Catalogue (EWC) and its National Implementation

The European Waste Catalogue (EWC), established by Commission Decision 2000/532/EC, serves as the harmonized reference point for waste classification across all EU member states.[1] It was designed to provide a common language for waste tracking, enabling consistent statistical reporting and transboundary shipment controls. In Germany, this catalogue is transposed into national law via the AVV.[3]

The significance of this for the "Fessi" chatbot is that the data structure is inherently multilingual and standardized. A waste code generated in a German university (e.g., *08 03 18* for waste toner) corresponds semantically to the same waste stream in France or Poland, facilitating potential cross-border scalability or the utilization of non-German training data to enhance the chatbot's natural language understanding (NLU) capabilities. The AVV is legally binding for the designation of waste and its classification as hazardous or non-hazardous.[3]

1.1.1 The Hierarchical Structure of the AVV

The AVV is not a flat list but a taxonomy organized by the *process of origin*. This "origin-based" logic is critical for the chatbot's decision tree. The catalogue is divided into 20 chapters (two-digit codes), which are further subdivided into groups (four-digit codes) and specific waste types (six-digit codes).[1]

- **Chapters 01 to 12:** Cover wastes from industrial processes, agriculture, and raw material processing. For example, Chapter 07 covers wastes from organic chemical processes, which is highly relevant for university chemistry departments.[5]
- **Chapters 13 to 15:** Cover wastes that are not process-specific but material-specific, such as waste oils (Chapter 13), solvents (Chapter 14), and packaging (Chapter 15).[6]
- **Chapter 16:** A "catch-all" chapter for wastes not otherwise specified in the list, including off-specification batches and laboratory chemicals. This chapter is of paramount importance for the university context.[7]
- **Chapters 17 to 19:** Cover construction and demolition waste (Chapter 17), healthcare waste (Chapter 18), and waste from waste management facilities (Chapter 19).[6]
- **Chapter 20:** Covers municipal wastes, including separately collected fractions from households and similar institutions like universities (e.g., paper, glass, bio-waste).[1]

The classification rules dictate a strict precedence: one must first attempt to classify waste under Chapters 01–12 or 17–20 based on the generating activity. Only if no suitable code is found there, should Chapters 13, 14, and 15 be considered. Chapter 16 is the final resort.[10] The "Fessi" knowledge graph must encode this procedural logic to prevent misclassification. For instance, a plastic bottle is *15 01 02* (packaging) if it was a container, but it might be *20 01 39* (plastics) if it was a non-packaging plastic item collected municipally, or *17 02 03* (plastic) if it

came from a construction site.

1.2 The Binary Logic of Hazardousness

A critical attribute in the AVV data model is the "hazardous" flag. Wastes considered hazardous are explicitly marked in the AVV with an asterisk (*) next to the six-digit code.[3] This binary distinction triggers vastly different legal obligations regarding storage, transport, and disposal evidence (*Nachweisverordnung*).

In the knowledge graph, isHazardous must be a core property of the WasteCode entity. This property drives the chatbot's safety warnings. If a user identifies a waste stream that maps to an asterisked code, "Fessi" must immediately trigger a "Danger" workflow, instructing the user on specific containment (e.g., "Use the red safety canister") and labeling requirements defined by TRGS 201.[11]

1.3 The Challenge of Mirror Entries

The most complex aspect of the AVV, and the one requiring the most sophisticated logic in the knowledge graph, is the concept of "mirror entries" (*Spiegeleinträge*). These are pairs (or sets) of codes describing the same type of waste, distinguished only by the concentration of dangerous substances.[4]

For example, construction soil can be classified as:

- **17 05 03*:** Soil and stones containing hazardous substances.
- **17 05 04:** Soil and stones other than those mentioned in 17 05 03.[13]

Similarly, laboratory chemicals mirror each other:

- **16 05 06*:** Laboratory chemicals, consisting of or containing hazardous substances.
- **16 05 09:** Discarded chemicals other than those mentioned in 16 05 06, 16 05 07 or 16 05 08.[14]

For the chatbot, a mirror entry represents a *decision node*. It cannot simply look up the code; it must prompt the user for qualitative or quantitative data regarding the waste's composition. "Does the soil contain oil or heavy metals?" or "Is the chemical toxic or flammable?" This requires the knowledge graph to link mirror entries to their specific "Hazardous Properties" (HP) criteria.

2. Classification Logic and Hazard Assessment

To automate the classification of mirror entries, "Fessi" requires access to the logic defined in the EU Waste Framework Directive (2008/98/EC) and the CLP Regulation (1272/2008/EC). The transition from the old "H-codes" (H1-H14) to the new "HP-codes" (HP 1 to HP 15) is a crucial

update that must be reflected in the system's reasoning engine.[4]

2.1 The HP Criteria (Hazard Properties)

The assessment of whether a waste is hazardous depends on whether it displays one or more of the properties HP 1 to HP 15. The knowledge graph must map these properties to user-understandable questions.

HP Code	Description	Logic for Chatbot	Data Source
HP 3	Flammable	Ask: "Is the liquid flammable? Does it have a flash point below 60°C?"	[16]
HP 4	Irritant	Ask: "Does the substance cause skin irritation or eye damage?"	[12]
HP 6	Acute Toxicity	Ask: "Is the substance fatal or toxic if swallowed/inhaled? " Check for skull & crossbones symbol.	[16]
HP 8	Corrosive	Ask: "Is the pH value ≤ 2 or ≥ 11.5?" (Young Method check required for extreme pH)	[12]
HP 14	Ecotoxic	Requires summation of concentrations of substances with aquatic toxicity (H400/H410).	[12]

Data Integration: The LAGA M31 and technical notes on waste classification provide the

specific concentration limits (e.g., total content of substances classified as H410 ≥ 0.25% triggers HP 14).[12] This mathematical logic must be embedded in the chatbot's backend or represented via SHACL (Shapes Constraint Language) rules within the graph.

2.2 Persistent Organic Pollutants (POPs)

A specialized subset of hazardous waste involves Persistent Organic Pollutants (POPs). The EU POPs Regulation (2019/1021) sets strict concentration limits for substances like HBCD (often found in insulation foam) or PCBs.[17]

If a waste contains POPs above the threshold (e.g., 1,000 mg/kg for HBCD), it is classified as hazardous and often requires destruction (usually incineration) rather than recycling.[19] The knowledge graph needs a distinct "POPs" flag for relevant waste streams (e.g., construction waste, plastics, WEEE) to ensure the chatbot advises against standard recycling paths for these materials.

2.3 The Role of TRGS 201

For universities, particularly those with chemical laboratories, the *Technische Regeln für Gefahrstoffe* (TRGS) 201 is the operational bible. It bridges the gap between the chemical law (GefStoffV) and waste law (KrWG).[11]

TRGS 201 enables a "simplified classification" for internal handling. It allows wastes to be grouped by hazard characteristics (e.g., "organic solvents, halogen-free") rather than requiring a detailed analysis of every single beaker. "Fessi" should utilize this rule set to suggest practical collecting categories to students. Instead of asking for the exact CAS number of every residue, the chatbot can ask, "Is it a halogenated solvent?" and map the answer to the group code *14 06 02** or *14 06 03**.[21]

3. Comprehensive Data Source Analysis

Populating the knowledge graph requires a multi-source ingestion strategy. No single dataset contains all the necessary layers of information (legal codes, synonyms, handling rules, and hazard logic).

3.1 Primary Legal Texts and Datasets

The authoritative source for the code list is the text of the AVV itself.

- **Gesetze-im-Internet (BMJ/Juris):** Provides the AVV in HTML and XML formats. This is the source of truth for the German descriptions.[3]
- **EU Open Data Portal & Eurostat:** Provides the EWC in RDF, XML, and bulk CSV formats.[23] The "Controlled Vocabularies" repository offers SKOS representations, which are natively compatible with knowledge graph technologies.[25] This source is essential for

obtaining official English translations if "Fessi" is to support international students.

3.2 The SEPA/UK Datasets (Structured Excellence)

While German sources provide the text, UK agencies (Environment Agency, SEPA, Natural Resources Wales) excel at providing *structured data*.

- **Consolidated Lists:** These agencies publish the "List of Wastes" in Excel/CSV formats that clearly separate "absolute" vs. "mirror" entries—a distinction often buried in the text of the German AVV.[1]
- **Waste Thesaurus:** SEPA publishes a "Waste Thesaurus".[1] This document maps thousands of common terms (e.g., "acetone," "pallets," "fluorescent tubes") to their EWC codes. This is arguably the most valuable dataset for the chatbot's Natural Language Understanding (NLU) layer, as users rarely search for "waste from organic chemical processes" but rather for "old paint" or "broken beaker."

3.3 Electronic Records (ZEDAL and eANV)

The electronic waste records procedure (*elektronisches Abfallnachweisverfahren* - eANV) is mandatory for hazardous waste in Germany.

- **ZEDAL:** As a leading provider of eANV software, ZEDAL maintains rigorous, validated catalogs of waste codes and disposal facility numbers.[28] For universities already using ZEDAL (which is common for hazardous waste management), there is an opportunity to align "Fessi's" internal IDs with the university's ZEDAL system, ensuring that the disposal pathways recommended by the chatbot match the active contracts in the administrative system.[29]

3.4 Unstructured University Repositories ("Abfall-ABC")

Every major German university publishes an "Abfall-ABC"—an alphabetical guide to waste disposal. These documents represent the "ground truth" of how waste is actually handled on specific campuses.

- **LMU Munich & Uni Heidelberg:** These institutions provide detailed PDFs linking specific lab wastes (e.g., Ethidium bromide, biological cultures) to internal disposal routes and AVV codes.[30]
- **ETH Zurich:** While Swiss, their guidelines on chemical compatibility and separation classes are extremely detailed and applicable to the German context due to similar safety standards.[32]
- **Data Extraction Strategy:** These sources are unstructured (PDF/HTML). Populating the knowledge graph requires scraping these documents to extract "Synonym -> Code" mappings. For example, mapping "Papiertücher" (Paper towels) to *18 01 04* (if from a hospital ward) or *20 03 01* (if from a washroom).

4. Domain-Specific Waste Streams in the University Context

A generic waste chatbot is insufficient for a university. The knowledge graph must be densely populated in specific chapters that reflect the unique activities of a research and education institution: Laboratory Science, Healthcare/Medicine, and Campus Infrastructure.

4.1 The Laboratory Domain (Chemicals and Solvents)

Laboratories are the primary source of hazardous waste on campus. The relevant AVV chapters are **06** (Inorganic), **07** (Organic), and **16** (Unspecified).

4.1.1 Solvent Classification (Chapter 14 vs. 07)

A common confusion arises with solvents.

- **14 06 03*:** Other solvents and solvent mixtures. This code is typically used for *pure* collected solvents (e.g., a canister of waste acetone from cleaning).[21]
- **07 01 04*:** Other organic solvents, washing liquids and mother liquors. This applies if the solvent waste arises directly from an organic synthesis process (MFSU).[21]
- **Graph Logic:** The chatbot must ask the user about the *origin*. "Is this solvent from a specific synthesis reaction (Chapter 07) or general cleaning/degreasing (Chapter 14)?" Most general lab solvent waste collected in "waste solvent canisters" is often classified under Chapter 14 or Chapter 16 in university practice.

4.1.2 Laboratory Chemicals (Group 16 05)

This is the "home" for most small-scale lab chemicals.

- **16 05 06*:** Laboratory chemicals, consisting of or containing dangerous substances, including mixtures.[14] This is the default for old reagents, reaction residues, and unknown mixtures.
- **16 05 09:** Discarded chemicals other than those mentioned in 16 05 06, 07 or 08. This is for non-hazardous salts (e.g., NaCl, Glucose).[34]

Table 1: Common Laboratory Chemical Mappings

Chemical / Waste Type	CAS Number	Potential AVV Codes	Hazard Classification
Acetone (Spent)	67-64-1	14 06 03* / 07 01 04*	HP 3 (Flammable)

Methanol	67-56-1	14 06 03* / 16 05 06*	HP 3 (Flammable), HP 6 (Toxic)
Sulfuric Acid	7664-93-9	06 01 01* / 16 05 06*	HP 8 (Corrosive)
Ethidium Bromide Gel	1239-45-8	16 05 06* / 18 01 03*	HP 7 (Carcinogenic), HP 11 (Mutagenic)
Solid Salts (Heavy Metal)	Various	16 05 07*	HP 6 (Toxic), HP 14 (Ecotoxic)
Solid Salts (Non-Haz)	Various	16 05 09	Non-Hazardous

4.2 The Healthcare and Biological Domain (Chapter 18)

For university hospitals and biology faculties, **LAGA Mitteilung 18** serves as the authoritative implementation guide.[35] The AVV divides this waste based on infection risk.

4.2.1 Infectious vs. Non-Infectious (The "B-Waste" Distinction)

- **18 01 03*:** Wastes whose collection and disposal is subject to special requirements in order to prevent infection. This includes cultures, infected blood, and waste from patients with highly contagious diseases (e.g., biologically active waste).
- **18 01 04:** Wastes whose collection and disposal is *not* subject to special requirements. This covers the vast majority of "hospital waste" like dressings, gowns, and swabs that are not specifically infectious. In Germany, this is often referred to as "B-Müll" and can often be incinerated with municipal waste (under strict hygiene controls).[35]
- **18 01 01 (Sharps):** Sharps (scalpels, needles) are collected separately. While legally they might fall under 18 01 01, university safety rules often mandate they be treated as infectious or hazardous to prevent injury, requiring specific "Sharps Containers".[37]

4.2.2 Cytostatics

A critical safety class is **18 01 08***: Cytotoxic and cytostatic medicines.[38] These are CMR substances (Carcinogenic, Mutagenic, Reprotoxic). The knowledge graph must contain a list of common cytostatic drugs (e.g., Cisplatin, Doxorubicin) and map them directly to this code to ensure they are never mixed with standard pharmaceutical waste (18 01 09).

4.3 Campus Infrastructure and Electronics (Chapters 17 & 20)

Construction and electronic waste (WEEE) are significant streams during renovations or equipment upgrades.

- **WEEE (Waste Electrical and Electronic Equipment):**
  - **20 01 35*:** Discarded electrical and electronic equipment containing hazardous components (e.g., CRT monitors, devices with batteries).[39]
  - **20 01 36:** Discarded electrical and electronic equipment other than those mentioned in 20 01 21, 20 01 23 and 20 01 35.[39]
  - **Data Security:** For a university, "Fessi" must add a layer of advice regarding data deletion for hard drives and devices, linking to the university's IT security policy.
- **Batteries:**
  - **20 01 33*:** Batteries and accumulators included in 16 06 01, 16 06 02 or 16 06 03 and unsorted batteries containing these batteries.
  - **Lithium-Ion Batteries:** These pose a fire risk. Fessi needs to provide specific instructions on taping terminals and using fire-safe collection boxes.[39]

5. Knowledge Graph Construction and Engineering

Transforming these diverse data sources into a functional system requires a formal ontology. The "Fessi" Knowledge Graph should be built using W3C standards (RDF, OWL, SKOS) to ensure interoperability and reasoning capabilities.

5.1 Ontology Schema Design

The schema must capture the relationships between codes, hierarchies, hazards, and handling rules.

- **Classes:**
  - avv:WasteCode: The central entity (e.g., "16 05 06*").
  - avv:Chapter, avv:Group: Hierarchical parents.
  - chem:Substance: Representing chemical entities (e.g., "Methanol").
  - hazard:Property: The HP criteria (e.g., "HP3").
  - proc:HandlingInstruction: Instructions like "Use Yellow Bin," "Do not compact."
  - reg:LegalSource: The document deriving the rule (e.g., "TRGS 201").
- **Properties:**
  - avv:hasCode: String (e.g., "160506").
  - avv:isHazardous: Boolean.
  - avv:mirrorOf: ObjectProperty linking mirror pairs.
  - chem:containsSubstance: Links waste streams to chemicals.
  - hazard:triggersCriteria: Links substances/wastes to HP codes.
  - skos:altLabel: Stores the thousands of synonyms from Abfall-ABCs (e.g., "Tipp-Ex", "Broken beaker").

5.2 Data Ingestion Strategy

1. **Skeleton Ingestion:** Parse the XML from *Gesetze-im-Internet* to create the 20 Chapters, Groups, and ~840 Waste Codes.
2. **Attribute Enrichment:** Join this data with the SEPA Consolidated List (CSV) to add robust "Absolute/Mirror" flags and English descriptions.[26]
3. **Synonym Injection:** Run a scraping pipeline on the targeted "Abfall-ABC" PDFs (LMU, Heidelberg, etc.). Use Named Entity Recognition (NER) to identify waste terms and the associated AVV codes in the documents. Populate skos:altLabel properties.
4. **Chemical Logic Mapping:** Ingest a "high-priority" chemical list (common lab reagents) with their CLP H-phrases. Use SHACL rules to map these to AVV codes (e.g., IF chem:hPhrase IN ["H225", "H226"] THEN suggestCode "14 06 03*").

5.3 Reasoning and Inference

The graph is not just a database; it is a reasoning engine.

- **Mirror Entry Resolution:** When a user queries a term that maps to a mirror entry (e.g., "Construction wood"), the graph does not return a single code. It identifies the mirrorOf relationship and triggers a dialogue: "Is the wood treated with hazardous preservatives?"
  - If YES -> Infers **17 02 04***.
  - If NO -> Infers **17 02 01**.
- **Hierarchy Traversal:** If a specific search (e.g., "HPLC Acetonitrile waste") fails, the graph can traverse up the hierarchy to Chapter 07 or 14 and suggest the general "Organic Solvents" code, implementing the "catch-all" logic of the AVV.

6. Conclusion and Strategic Roadmap

The development of "Fessi" represents a critical step in modernizing university waste management. By moving from static PDF lists to a semantic knowledge graph, the system can provide context-aware, legally compliant advice that adapts to the specific chemical and biological hazards present on a campus.

The roadmap for data sourcing is clear:

1. **Anchor the system** in the official AVV text from *Gesetze-im-Internet*.
2. **Enrich the structure** using the structured datasets from SEPA and Eurostat to handle mirror entries and multilingual support.
3. **Localize the knowledge** by ingesting "Abfall-ABC" data from peer institutions like LMU and ETH Zurich to capture the vernacular of the university environment.
4. **Operationalize the logic** by encoding the decision trees of LAGA M18 (healthcare) and TRGS 201 (chemicals) directly into the graph's inference layer.

This architecture ensures that "Fessi" is not merely a search engine for codes, but an intelligent compliance assistant capable of navigating the complex intersection of chemical

safety, environmental law, and campus logistics.

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